Cluster 17677

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Subjects

Research fields: Computer science, Engineering, Mathematics
The most common research fields of articles in this cluster. Read more.

Research subfields: Computer engineering, Electronic engineering, Algorithm
Common research subfields of articles in this cluster. Read more.

Key concepts: Intelligent Reflecting Surface, Reconfigurable Intelligent Surfaces, wireless communication systems, wireless networks, RIS Assisted Wireless
These key concepts are identified algorithmically using article titles and abstracts. Read more.

2.14% of all articles are AI-related
Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.13% of English-language articles are related to robotics
Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.04% of English-language articles are related to computer vision
Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.01% of English-language articles are related to natural language processing
Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.04% of English-language articles are related to AI safety
Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

1.57% of English-language articles are related to cybersecurity

Vital signs

9479 recent articles in the cluster
9479 articles in this cluster were published in the five year period ending in 2023.

Average article is **3.12** years old
The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **87.32th** percentile
Over the three year period ending in 2023, this cluster grew faster than roughly 87.32% of other clusters.

Citation rating: **60.02th** percentile
A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year.In this case, the average article in this cluster is cited more often than roughly 60.02% of all other articles published in the same year.

96.82% of articles are in English

Expected to grow extremely fast
As determined by ETO's growth prediction model. Read more.

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 8619 of 9479 articles. 7181 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	4404	4.45
United Kingdom	1272	6.80
United States	953	6.89
India	809	2.76
Canada	560	6.67
South Korea	556	4.37
Singapore	411	10.63
France	322	10.15
Australia	321	7.54
Germany	316	8.62

Top funding countries

Data available for 4596 of 9479 articles. 4006 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	2977	5.33
United States	401	8.15
South Korea	297	4.48
United Kingdom	259	9.06
Belgium	230	14.31
Singapore	171	10.56
Canada	141	6.80
Australia	98	9.64
Taiwan	92	4.78
Germany	86	8.93

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

Reconfigurable Intelligent Surface Assisted Multiuser MISO Systems Exploiting Deep Reinforcement Learning
2020: arXiv: Information Theory. 752 citations.

Active RIS vs. Passive RIS: Which Will Prevail in 6G?
2021: arXiv: Information Theory. 547 citations.

An Overview of Signal Processing Techniques for RIS/IRS-Aided Wireless Systems
2021: arXiv. 353 citations.

Simultaneously Transmitting and Reflecting (STAR) RIS Aided Wireless Communications
2021: arXiv: Information Theory. 433 citations.

Smart Radio Environments Empowered by Reconfigurable Intelligent Surfaces: How It Works, State of Research, and The Road Ahead
2020: IEEE Journal on Selected Areas in Communications. 2205 citations.

A Survey on Channel Estimation and Practical Passive Beamforming Design for Intelligent Reflecting Surface Aided Wireless Communications
2021: arXiv: Information Theory. 319 citations.

Reconfigurable Intelligent Surfaces: Principles and Opportunities
2020: arXiv: Signal Processing. 990 citations.

Channel Estimation for RIS-Empowered Multi-User MISO Wireless Communications
2020: arXiv: Information Theory. 538 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

Review on Channel Estimation for Reconfigurable Intelligent Surface Assisted Wireless Communication System
2023: Mathematics. 3 citations.

IRS-Empowered 6G Networks: Deployment Strategies, Performance Optimization, and Future Research Directions
2022: IEEE Access. 16 citations.

A Survey on STAR-RIS: Use Cases, Recent Advances, and Future Research Challenges
2023: IEEE Internet of Things Journal. 69 citations.

Review on Channel Estimation of Reconfigurable Intelligent Surface Assisted Wireless Communication System
2023. 3 citations.

An Overview of Signal Processing Techniques for RIS/IRS-Aided Wireless Systems
2021: arXiv. 353 citations.

A Survey on Channel Estimation and Practical Passive Beamforming Design for Intelligent Reflecting Surface Aided Wireless Communications
2021: arXiv: Information Theory. 319 citations.

Deep Learning Aided Intelligent Reflective Surfaces for 6G: A Survey
2024: ACM Computing Surveys. 0 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

Reconfigurable Intelligent Surface Assisted Multiuser MISO Systems Exploiting Deep Reinforcement Learning
2020: arXiv: Information Theory. 752 citations.

Active RIS vs. Passive RIS: Which Will Prevail in 6G?
2021: arXiv: Information Theory. 547 citations.

6G Internet of Things: A Comprehensive Survey
2021: arXiv. 651 citations.

The Road Towards 6G: A Comprehensive Survey
2021: arXiv. 998 citations.

Toward 6G Networks: Use Cases and Technologies
2020: IEEE Communications Magazine. 1493 citations.

Towards 6G wireless communication networks: vision, enabling technologies, and new paradigm shifts
2020: 中国科学. 信息科学. 1515 citations.

Smart Radio Environments Empowered by Reconfigurable Intelligent Surfaces: How It Works, State of Research, and The Road Ahead
2020: IEEE Journal on Selected Areas in Communications. 2205 citations.

Reconfigurable Intelligent Surfaces: Principles and Opportunities
2020: arXiv: Signal Processing. 990 citations.

Channel Estimation for RIS-Empowered Multi-User MISO Wireless Communications
2020: arXiv: Information Theory. 538 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 9357 out of 9479 articles. 3582 of these articles came from these sources.

Source	Articles (last five years)
arXiv (Cornell University)	1346
arXiv.org	325
IEEE Transactions on Vehicular Technology	316
IEEE Wireless Communications Letters	294
arXiv: Information Theory	289
IEEE Communications Letters	232
IEEE Transactions on Wireless Communications	230
IEEE Global Communications Conference	204
IEEE Internet of Things Journal	175
arXiv	171

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 9401 of 9479 articles. 1162 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Qingqing Wu	199	27.13
Marco Di Renzo	171	76.33
Yuanwei Liu	161	36.46
Chau Yuen	157	34.94
Cunhua Pan	144	24.40
Shi Jin	130	20.70
George C. Alexandropoulos	126	32.99
Zhu Han	114	30.90
Chongwen Huang	107	47.73
Jiangzhou Wang	101	12.94

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 8658 of 9479 articles. 2251 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Southeast University - China	564	6.06
University of Electronic Science and Technology of China	357	5.46
Beijing University of Posts and Telecommunications - China	344	6.49
Nanjing University of Posts and Telecommunications - China	269	3.1
Queen Mary University of London - United Kingdom	258	9.73
Xidian University - China	219	5.14
Nanyang Technological University - Singapore	211	7.51
Zhejiang University - China	193	8.02
Shanghai Jiao Tong University - China	190	5.28
Tsinghua University - China	182	6.94

Organization types

This table covers the 8314 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 9479 articles total). Read more

Commercial	Education	Nonprofit	Government
2.43%	46.85%	3.80%	1.23%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 729 out of 9479 articles. 353 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
China Mobile (China)	95	8.01
ZTE (China)	68	7.16
Huawei Technologies (China)	56	11.82
China Telecom - China	47	1.97
Ericsson (Sweden)	30	5.78
Orange (France)	21	6.63
NTT (Japan)	20	1.03
Samsung (South Korea)	20	2.03
Nokia (Finland)	18	5.95
InterDigital (United Kingdom)	17	5.32

Top funders

Data available for 5408 out of 9479 recent articles. 3591 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	2413	5.59
Ministry of Science and Technology - China	626	7.09
Ministry of Education - China	557	6.97
National Key Research & Development Program of China	482	8.37
National Science Foundation - United States	315	8.98
Engineering & Physical Sciences Research Council (EPSRC)	189	12.11
European Union	181	4.75
Engineering and Physical Sciences Research Council - United Kingdom	178	10.71
National Research Foundation of Korea - South Korea	174	5.03
European Commission - Belgium	169	14.33

Funder types

This table covers the 4243 articles in this cluster from the last five years with data about their funding organization types (out of 9479 articles total).

Commercial	Education	Nonprofit	Government
2.10%	9.74%	8.50%	79.66%

Patents and industry

913 patents cite articles in this cluster (**98.89th** percentile among all clusters)

7.34% of articles in the cluster are cited by patents

2.3% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
CHANNEL ESTIMATION TECHNIQUE FOR RECONFIGURABLE INTELLIGENT SURFACES	24
RECONFIGURABLE INTELLIGENT SURFACE ENABLED LEAKAGE SUPPRESSION AND SIGNAL POWER MAXIMIZATION SYSTEM AND METHOD	18
GRID OF BEAM -TYPE DESIGN AND IMPLEMENTATION OF A RE-CONFIGURABLE INTELLIGENT SURFACE	18
SURFACE FOR CONTROLLED RADIO FREQUENCY SIGNAL PROPAGATION	14
COVERT COMMUNICATION TECHNIQUE FOR INTELLIGENT REFLECTING SURFACE ASSISTED WIRELESS NETWORKS	13
METHOD AND DEVICE FOR ENHANCING POWER OF SIGNAL IN WIRELESS COMMUNICATION SYSTEM USING IRS	13
LARGE INTELLIGENT SURFACES WITH SPARSE CHANNEL SENSORS	11
WIRELESS TELECOMMUNICATIONS NETWORK	11
INITIALIZATION AND OPERATION OF INTELLIGENT REFLECTING SURFACE	11

Top patent assignees

Patent assignee	Same-cluster citations
QUALCOMM INCORPORATED	79
ERICSSON (SWEDEN)	62
HUANG SHUYING DR	60
NISSAN (JAPAN)	47
ZHANG YU	46
UNIV NANJING POSTS & TELECOMMUNICATIONS	44
NANJING AEROSPACE UNIVERSITY	43
SONOVA (UNITED STATES)	41
QUADRATURE ZHISHI MEDICINE CO LTD	102

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
12959	283	Integrated Sensing, ISAC systems, Communication Integrated System, ISAC waveform design, sensing performance
660	198	unmanned aerial vehicles, Mobile Edge Computing, UAV trajectory, UAV communications, UAV networks
8070	169	wireless communication systems, Terahertz Wireless Communications, THz wireless systems, THz channel, channel model
54098	161	Cell-Free Massive MIMO, Massive MIMO Systems, massive Multiple-input multiple-output, MIMO Networks, Uplink Cell-Free Massive
412	148	massive MIMO systems, Channel Estimation, mmWave massive MIMO, beam training, mmWave MIMO channel
18995	147	Ambient Backscatter Communications, Backscatter NOMA Systems, multiple backscatter devices, Cooperative Ambient Backscatter, Backscatter Cooperative Communication
724	133	optical encryption, metasurface optical elements, metasurface holography, metasurface polarization imaging, multifunctional metasurfaces
1453	129	non-orthogonal multiple access, Multiple Access, NOMA Systems, power allocation, NOMA Networks
9948	127	Federated Learning, machine learning models, distributed machine learning, Privacy-preserving Federated Learning, training data
11206	118	LEO satellite communication, LEO Satellite, Satellite Terrestrial Networks, communication networks, Satellite Systems

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
412	561	massive MIMO systems, Channel Estimation, mmWave massive MIMO, beam training, mmWave MIMO channel
660	472	unmanned aerial vehicles, Mobile Edge Computing, UAV trajectory, UAV communications, UAV networks
1453	432	non-orthogonal multiple access, Multiple Access, NOMA Systems, power allocation, NOMA Networks
926	271	Energy Harvesting, Wireless Energy Transfer, Wireless Sensor Networks, simultaneous wireless information, Power Transfer
724	259	optical encryption, metasurface optical elements, metasurface holography, metasurface polarization imaging, multifunctional metasurfaces
1897	246	Mobile Edge Computing, Edge Computing Networks, computation offloading, Deep Reinforcement Learning, resource allocation
8070	229	wireless communication systems, Terahertz Wireless Communications, THz wireless systems, THz channel, channel model
12959	220	Integrated Sensing, ISAC systems, Communication Integrated System, ISAC waveform design, sensing performance
732	218	Massive MIMO Systems, FDD Massive MIMO, channel estimation, MIMO Channel, massive Multiple-Input Multiple-Output
254	172	Physical Layer Security, secrecy outage probability, secrecy performance, channel state information, Cognitive Radio Networks

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
emilbjornson/SP_Cup_2021	3	41
marcantonio14/RIS-Power-Measurements-Dataset	2	2
Brook1711/RIS_components	2	63
recepakiftasci/Amplifying-RIS	2	0
baturaysaglam/RIS-MISO-Deep-Reinforcement-Learning	2	138
bilepeng/risnet	2	10
lucasanguinetti/Holographic-MIMO-Small-Scale-Fading	2	11
RujingXiong/RIS_Optimization	2	29
XML124/CDRN-channel-estimation-IRS	2	46
thanhluannguyen/UAV-RIS-blockage	1	7